

Linux (Operating System Kernel)

Linux is an open-source **operating system kernel**, which means it is the core part of an operating system that directly interacts with computer hardware. It was created by Linus Torvalds in 1991. Linux manages system resources like CPU, memory, and devices, and allows software to run efficiently. Being open-source, anyone can view, modify, and distribute its code. Linux is widely used in servers, mobile devices (like Android), supercomputers, and embedded systems due to its security, stability, and flexibility.

Ubuntu (Linux Distribution)

Ubuntu is a popular **Linux distribution (distro)** based on the Linux kernel. It is developed by Canonical Ltd.. Ubuntu provides a complete operating system by combining the Linux kernel with software like a graphical interface, applications, and package management tools. It is user-friendly, making it suitable for beginners as well as professionals. Ubuntu is widely used for desktops, servers, and cloud computing because it is free, secure, and regularly updated.

Linux – Features & Advantages

Key Features of Linux

1. **Open Source**
Linux is open-source, meaning its source code is freely available and can be modified by anyone.
2. **Multiuser System**
Multiple users can access the system at the same time without interference.
3. **Multitasking**
Linux can run multiple processes simultaneously (e.g., browsing + coding + music).
4. **Security**
Strong user permissions and authentication make it highly secure against viruses.
5. **Portability**
Linux can run on different hardware platforms like computers, servers, and embedded devices.
6. **Stability & Reliability**
Linux systems rarely crash and can run for long periods without reboot.
7. **Command Line Interface (CLI)**
Powerful terminal commands allow advanced control over the system.

Advantages of Linux

- **Free of Cost** – No license required.
- **Highly Customizable** – Users can modify according to their needs.

- **Better Performance** – Works well even on old hardware.
- **Strong Community Support** – Large global developer community.
- **Less Virus Attacks** – More secure than many other OS.

Ubuntu – Features & Advantages

Key Features of Ubuntu

1. **User-Friendly Interface**
Ubuntu provides a simple and attractive GUI, easy for beginners.
2. **Regular Updates**
Developed by Canonical Ltd., it receives frequent updates and security patches.
3. **Software Center**
Easy installation/removal of applications through Ubuntu Software Center.
4. **Pre-installed Applications**
Comes with a browser, office tools, media player, etc.
5. **Strong Security**
Built-in firewall and virus protection features.
6. **LTS (Long Term Support)**
Provides long-term updates (usually 5 years).
7. **Compatibility**
Supports a wide range of hardware and software.

Advantages of Ubuntu

- **Beginner Friendly** – Easy to install and use.
- **Free & Open Source** – No cost for usage.
- **Stable & Secure** – Ideal for personal and professional use.
- **Large Community Support** – Easy to find help online.
- **Wide Application Support** – Thousands of free applications available.

Ubuntu Directory Commands:-

Directory commands are used to **create, view, navigate, and manage folders (directories)** in Ubuntu/Linux. These commands are executed in the **Terminal (CLI)**.

1. pwd (Print Working Directory)

Theory

This command shows the **current directory path** where you are working.

Syntax

`pwd`

Example

`pwd`

Output

`/home/rahul/Documents`

Explanation

It tells you your current location in the file system.

2. ls (List Directory Contents)

Theory

Used to **display files and directories** inside a folder.

Syntax

`ls`

`ls -l`

`ls -a`

Example

`ls`

Output

`file1.txt folder1 image.png`

Explanation

- `ls` → shows basic list
- `ls -l` → detailed list
- `ls -a` → shows hidden files

3. cd (Change Directory)

Theory

Used to **move from one directory to another**.

Syntax

```
cd directory_name
```

Examples

```
cd Documents
```

```
cd ..
```

```
cd /
```

Explanation

- `cd Documents` → go inside folder
- `cd ..` → move one step back
- `cd /` → go to root directory

4. mkdir (Make Directory)

Theory

Creates a **new directory (folder)**.

Syntax

```
mkdir folder_name
```

Example

```
mkdir Projects
```

Explanation

Creates a new folder named *Projects*.

5. rmdir (Remove Directory)

Theory

Deletes an **empty directory only**.

Syntax

```
rmdir folder_name
```

Example

```
rmdir Projects
```

Explanation

Removes the folder if it is empty.

6. rm -r (Remove Directory with Files)

Theory

Used to **delete a directory along with its contents**.

Syntax

```
rm -r folder_name
```

Example

```
rm -r Projects
```

Explanation

Deletes the folder and all files inside it.

 *Use carefully (data cannot be recovered easily).*

7. cp -r (Copy Directory)

Theory

Copies a directory and its contents to another location.

Syntax

```
cp -r source destination
```

Example

```
cp -r Projects Backup
```

Explanation

Copies *Projects* folder into *Backup*.

8. mv (Move/Rename Directory)

Theory

Used to **move or rename a directory**.

Syntax

```
mv old_name new_name
```

Examples

```
mv Projects NewProjects  
mv Projects /home/rahul/Desktop
```

Explanation

- Rename folder
- Move folder to another location

Ubuntu File Commands:-

File commands are used to **create, view, copy, move, delete, and manage files** in Ubuntu/Linux using the terminal.

1. touch (Create File)

Theory

Creates a **new empty file**. If the file already exists, it updates its timestamp.

Syntax

```
touch filename
```

Example

```
touch file1.txt
```

Explanation

Creates a new file named *file1.txt* in the current directory.

2. cat (View/Create File Content)

Theory

Used to **display file content**, and also to create files with content.

Syntax

```
cat filename
```

Example

```
cat file1.txt
```

Explanation

Displays the content of the file on the terminal.

👉 To create file with content:

```
cat > file1.txt
```

3. nano (Edit File)

Theory

A simple text editor used to **create and edit files**.

Syntax

```
nano filename
```

Example

```
nano file1.txt
```

Explanation

Opens file in editor where you can type and save content.

4. cp (Copy File)

Theory

Used to **copy files** from one location to another.

Syntax

`cp source destination`

Example

`cp file1.txt file2.txt`

Explanation

Creates a copy of *file1.txt* named *file2.txt*.

5. mv (Move/Rename File)

Theory

Used to **move or rename files**.

Syntax

`mv oldname newname`

Examples

`mv file1.txt newfile.txt`

`mv file1.txt /home/rahul/Desktop`

Explanation

- Renames file
- Moves file to another directory

6. rm (Remove File)

Theory

Deletes a file permanently.

Syntax

rm filename

Example

rm file1.txt

Explanation

Removes the file from the system.

 Deleted files cannot be easily recovered.